Junsang Song

PERSONAL DETAILS

Name:	Junsang Song
Designation:	Automotive Developer (Intern)
Date of Birth (yyyy/mm/dd):	2000/05/19
Email:	JunsangS@fpt.com

EDUCATIONAL BACKGROUND

- Incheon National University, 03/2019 – 08/2026
- 42Seoul, 03/2023 – 08/2024

CERTIFICATES

- Github/perman0519

PROFESSIONAL SUMMARY

- Proficient in **C/C++ with POSIX and STL**
- Experienced in **ROS2** and **Autonomous Driving** including winning the Grand Prize at the KAIST Mobility Challenge
- **Linux** and **Docker** experience
- **AUTOSAR** Training via Inflearn
- Familiar with **Agile/Scrum**

TECHNICAL SKILLS

Skill	Proficiency	Experience
Language		
C/C++	Intermediate	2y
Python	Fundamental	1y
System Programming		
POSIX	Fundamental	1y
Memory Management	Fundamental	1y
Framework/Library		

ROS2	Fundamental	1y
Tools		
Git	Intermediate	3y
vscode	Fundamental	3y
Bash	Fundamental	2y
Docker	Fundamental	1y
Operating Systems		
Linux	Intermediate	3y
Windows	Intermediate	5y
MacOS	Intermediate	5y

LANGUAGE SKILLS

- English Toeic Speaking IH

PROFESSIONAL EXPERIENCE

- Total work experience (Year/Month): 0

PROJECT EXPERIENCE

Project 1 V2X Cooperative Autonomous Driving 2025 KAIST Mobility Challenge	
Duration	11/2025 - 02/2026
Team Size	6
Position	Team leader
Description	By improving the decision-making and control logic for V2X autonomous driving, lane departures were reduced from 15 to 1, and vehicle collisions were reduced from 5 to 1.
Country/Industry	Korea/KAIST
Business Domain	Autonomous Driving
Responsibility	• Lead team • Development of a Rule-Based Algorithm using C++ and ROS2. • Development of a Python visualization program for lane visualizer • Reducing code upload time for 4 vehicles using SSH and git branch • Code management using Git/Github
Technology	• ROS2

	• C++ • Python • Git • Docker • Autonomous Driving • Linux

Project 2 Autonomous Driving 2025 student_mobility_competition	
Duration	04/2023 - 09/2024
Team Size	12
Position	Autonomous Driving SW Developer (ROS2)
Description	Improved cone-based path planning, increasing path generation success rate from 10% to 95% by analyzing and adapting open-source algorithms.
Country/Industry	Korea/Automotive
Business Domain	Automotive
Responsibility	• Developed GPS/IMU sensor fusion system • Developed cone-based path planning algorithm • Researched path planning algorithms and analyzed open-source implementations from Formula Student Germany teams
Technology	• ROS2 • Python • Git/Github

Project 3 Remote Smart Farm ESP32 + MQTT	
Duration	08/2024 - 11/2024
Team Size	4
Position	Embedded Developer
Description	Reduced remote control latency from 5s to 0.5s by building an MQTT-based real-time data pipeline on ESP32
Country/Industry	Korea/Smart Farm
Business Domain	Smart Farm
Responsibility	• Developed ESP32-based sensor data acquisition and control logic • Designed MQTT-based real-time communication pipeline for remote control • Implemented message structure and topic architecture for stable data transmission

	• Handled communication failures with reconnection and timeout logic
Technology	• C (Embedded programming) • ESP32, ATmega128p • MQTT Protocol • JavaScript (web interface) • webOS (application deployment)

Project 4 mini Shell (Bash-like)	
Duration	05/2023 - 08/2023
Team Size	4
Position	System Software Developer (C / POSIX)
Description	Implemented core Bash functionalities with zero memory leaks by designing fork/exec-based pipelines and file descriptor close/wait rules.
Country/Industry	Seoul/SW
Business Domain	SW
Responsibility	• Designed and implemented fork/exec-based process pipelines • Managed file descriptor (FD) lifecycle with close/wait rules to prevent resource leaks • Implemented command parsing and execution flow for Bash-like shell • Debugged memory issues and ensured zero memory leaks through systematic testing
Technology	• C (System programming) • POSIX (fork, exec, pipe, wait) • Bash • Makefile

Project 5 MiniRT - Ray Tracing	
Duration	01/2024 - 03/2024
Team Size	2
Position	Graphics / Rendering Developer (C)
Description	Reduced 2K rendering time by 4 by applying multithreading with tile-based workload partitioning.
Country/Industry	Korea/SW
Business Domain	SW
Responsibility	• Implemented ray tracing rendering pipeline in C

	• Designed tile-based workload partitioning for parallel processing • Applied multithreading to distribute rendering tasks across CPU cores • Optimized rendering performance and reduced execution time by 4×
Technology	• C (System / graphics programming) • Ray Tracing • miniLibX • Git • Bash